

Muskits-ESPnet: A Comprehensive Toolkit for Singing Voice Synthesis in New Paradigm

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Abstract

This research presents Muskits-ESPnet, a versatile toolkit that introduces new paradigms to Singing Voice Synthesis (SVS) through the application of pretrained audio models in both continuous and discrete approaches. Specifically, we explore discrete representations derived from SSL models and audio codecs and offer significant advantages in versatility and intelligence, supporting multi-format inputs and adaptable data processing workflows for various SVS models. The toolkit features automatic music score error detection and correction, as well as a perception auto-evaluation module to imitate human subjective evaluating scores. Muskits-ESPnet is available at <https://github.com/espnet/espnet>.

CCS Concepts

• Applied computing → Sound and music computing.

Keywords

Singing Voice Synthesis, Pretrained Model, Music Processing

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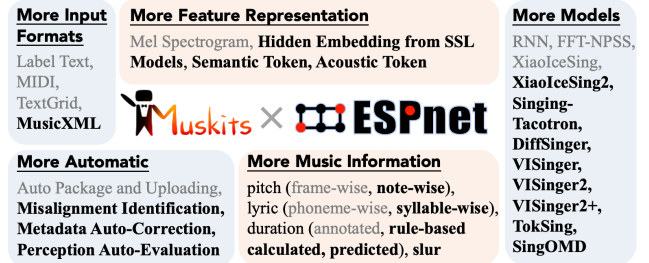


Figure 1: Improvements of Muskits-ESPnet compared with its origin version. The boldface indicates new functions.

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1 Introduction

SVS converts music scores into vocal singing using a specific singer's voice, aiming for accurate lyrics, pitch, and duration while ensuring a realistic sound. It faces challenges in achieving high standards of pitch, prosody, and emotional expression due to complex data processing requirements.

The common approach for SVS [12, 13, 21, 25, 32] involves an acoustic model predicting acoustic feature representations from music scores, followed by a vocoder [11, 16, 17, 22] reconstructing audio from these features. Most music processing toolkits [23, 38] for SVS, including our initial version of Muskits [26], follow this framework. However, the emergence of audio pre-training and the shift towards discrete representations in large models have brought new possibilities for SVS. Previously, 80-dimensional real-valued mel-spectrograms are commonly used as acoustic representations. Now, outputs from audio pretrained models [2, 3, 9, 15, 19, 28] trained on

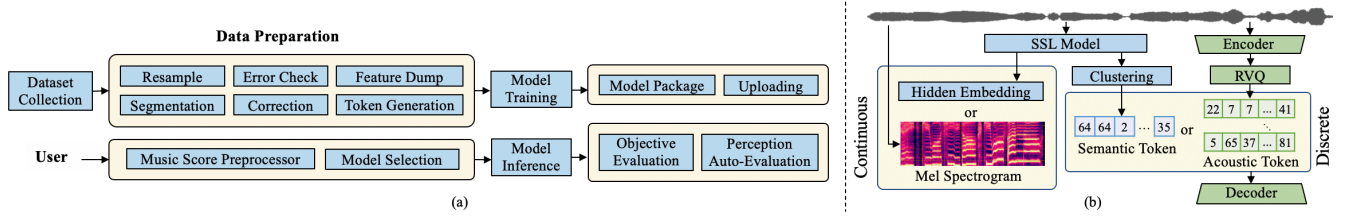


Figure 2: (a) SVS workflow. The upper section illustrates the training pipeline, while the lower section shows the inference pipeline for users. The functions in the yellow blocks can be flexibly selected based on specific requirements. **(b) Different SVS feature representations.** Continuous features include mel spectrograms and hidden embeddings from SSL models. Discrete representations consist of semantic tokens clustered from SSL models and acoustic tokens extracted from codecs.

large-scale datasets can assist acoustic modeling or extract discrete representations [4, 7, 8, 14, 18, 24, 27–29, 39, 40]. This approach efficiently meets the need for data discretization with large models [1, 31, 44]. Our work focuses on these new SVS paradigms and optimizes the entire data flow accordingly.

Our Muskits-ESPnet toolkit demonstrates exceptional versatility and intelligence (see Figure 1). We enhance the SVS models by integrating pre-trained models with traditional continuous feature-based approaches and introducing a new paradigm based on discrete representations. Furthermore, the entire data processing workflow is optimized to support all music file formats, not just specific datasets, and includes an automatic error-check and correction module to improve data alignment accuracy. We compile common feature representations to accommodate different SVS model inputs and introduce a perception auto-evaluation model [42], significantly reducing the cost and effort of manual scoring. Our Muskits-ESPnet toolkit supports the most advanced SVS models and automates the entire data processing workflow (see Figure 2). Recently, our toolkit serves as the baseline for the SVS track in Interspeech 2024 Discrete Speech Unit Challenge [6].

2 New Paradigms in SVS

Advances in audio pretraining technology impact audio generation tasks significantly. We apply this to SVS in two ways:

First, we enhance traditional SVS models by integrating pre-trained audio encodings, replacing or complementing mel spectrograms (see Figure ??). Our new SVS model [41], based on a Variational Auto-Encoder [46], performs better with joint encoding than with spectrograms alone.

Second, we explore SVS using discrete representations from pre-trained models, including semantic tokens from SSL model outputs and acoustic tokens from an audio codec [10, 43]. Our discrete-based SVS models [30, 37] in ESPnet achieve lower spatial costs compared to continuous representations.

3 Implementations

The Muskits-ESPnet data flow, illustrated in Figure 2, includes resampling, segmenting, error correction, and feature computation during training. Post-training, the model is packaged for upload. For inference, user inputs are preprocessed, the SVS model is selected, and evaluations are performed. Our framework supports various data types and model configurations, offering flexible functionality based on user needs. Detailed procedures for data preparation, training, inference, and evaluation are provided.

3.1 Data Preparation

This stage involves preprocessing raw music data into input sequences for SVS models. Typically, we extract sequences of three essential elements: <lyrics, pitch, duration> from various formats such as MusicXML, MIDI, and TextGrid. Upon reviewing several datasets, we identify a notable percentage of annotation errors, including redundant, missing, or misalignment of lyrics and notes. To tackle these issues, we develop a misalignment detection module and a metadata auto-correction module with specific adaptations for different languages. Our toolkit ensures annotation alignment consistency, thereby significantly enhancing model performance [35].

3.2 Training and Inference

Model training and inference follow the ESPnet [34] task processing workflow, supporting multi-GPU training and dynamic batching. We have significantly enhanced the generalizability of learning methods for SVS. This includes enriching joint training and fine-tuning paradigms for acoustic models and vocoders [36], and supporting both autoregressive [33] and non-autoregressive [5, 20, 21, 25, 30, 32, 37, 41, 45, 46] acoustic prediction methods. Additionally, the vocoder section now accommodates both continuous and discrete representations and includes transfer learning workflows [30, 37]. We have also optimized the data processing workflow, ensuring compatibility with different models while reducing time costs by approximately 60% compared to the previous generation.

3.3 Evaluation

We employ a comprehensive set of objective metrics to evaluate the similarity between generated audio and the original audio across various dimensions, including Mel Cepstral Distortion (MCD), Root Mean Square Error of Fundamental Frequency (F0_RMSE), Semitone Accuracy (SA), and Voiced/Unvoiced Error Rate (VUV_E). For listening feelings, we introduce an innovative perception auto-evaluation module [42] to emulate human judgment.

4 Conclusion

Muskits-ESPnet advances SVS by integrating audio pretraining and exploring both continuous and discrete representations, enhancing model capability and efficiency. It features robust data preprocessing, error correction, and support for diverse inputs. Optimized training and inference workflows, along with auto-evaluation, demonstrate its potential to support cutting-edge SVS models while reducing costs, setting a new standard for future SVS developments.

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